

- 7 (a) A photon has an energy of $3.11 \times 10^{-19} \text{ J}$.

Calculate the momentum of the photon.

momentum = N s [2]

- (b) A laser beam has a power of 350 mW. The light from the laser has a wavelength of 640 nm.

- (i) Determine the number of photons emitted by the laser in a time of 1.0 s.

number = [2]

- (ii) The laser beam is incident normally on a surface that absorbs all of the photons.

Show that the force F exerted on the surface by the laser beam is given by

$$F = \frac{P}{c}$$

where P is the power of the laser beam and c is the speed of light.

[2]

- (c) Light of a single wavelength is incident on the surface of different metals. The work function energy of the metals is given in Table 7.1.

Table 7.1

metal	work function energy / eV
tungsten	4.49
magnesium	3.68
potassium	2.26

- (i) Explain the term threshold wavelength.

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 [1]

- (ii) For the metals in Table 7.1, calculate the value of the largest threshold wavelength.

threshold wavelength = m [2]

[Total: 9]